

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

Linrui Xu^a, Linrui Han^{b,c,*}

^a*University of Winnipeg, Winnipeg, Manitoba, Canada*

^b*Ministry of Education Laboratory of Philosophy and Social Sciences – The CUPL Data Law Lab, China University of Political Science and Law, Beijing, China*

^c*The Institute for Data Law, China University of Political Science and Law, Beijing, China*

Abstract

Recovering prior client-specific advice from a legal consultation archive is difficult because dense retrieval may rank a relevant question turn highly while omitting the lawyer’s answer—an *episode incompleteness* failure under same-domain interference. BIMS-LEGAL is a dual-store system for Chinese legal consultation logs: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 performs turn-level session binding by giving siblings attenuated inherited scores (β s) rather than hard score copy. We evaluate exact replay as diagnostic and use paraphrase, follow-up, and advice-recall as primary non-exact channels. BM25-joint is a strong lexical baseline on exact and high-overlap queries, but Soft O2 is more effective on non-exact channels where online turn-level retrieval can recover a partial episode and then bind its answer sibling. Soft O2-C and gated Hybrid are reported as exploratory, gold-assisted solvability bounds for cross-session recovery, not as claims about natural unsupervised deployment. We report corrected graded nDCG alongside Answer Hit and Episode Completeness, with paired significance tests and a failure taxonomy separating complete, incomplete, and missed-session retrieval.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

*Corresponding author.

Email addresses: xu-l81@webmail.uwinnipeg.ca (Linrui Xu),
linrui_han@163.com (Linrui Han)

1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL is a dual-store Brain-Inspired Memory System for legal consultation archives, designed and evaluated in this study. The *systemic contribution* is a dual-store architecture that separates a fast episodic pathway (timestamped turn embeddings with speaker role and session identifier *sid*) from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) supplies an architectural analogy for this dual-pathway design, not an empirical claim about biological memory. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top- k —a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

36 The *algorithmic contributions* are soft binding operators that complete
 37 episodes without hard score copy:

- 38 • **Soft O2 (session soft binding).** When a turn is retrieved with score
 39 s , siblings sharing *sid* inherit $\max(s_{\nu}, \beta \cdot s)$, surfacing answer turns
 40 under question-hit / answer-miss while preserving rank competition
 41 among candidates.
- 42 • **Soft O2-C and gated Hybrid (cluster soft binding).** The same
 43 soft-inheritance rule applies within BIRCH clusters when gold evidence
 44 is not co-session with the query; Hybrid triggers cluster binding only
 45 on direct dense hits to avoid displacing same-session siblings.

46 Two **implementation and scale design operators** support rigorous large-
 47 scale shared-store experiments but are not standalone algorithmic contribu-
 48 tions:

- 49 • **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns,
 50 IVFPQ product quantization is under-trained and collapses Answer
 51 Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted
 52 sibling signals.
- 53 • **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes
 54 make archive construction tractable without changing retrieval seman-
 55 tics.

56 We ask four research questions:

- 57 • **RQ1.** Under same-domain interference, does episode incompleteness
 58 (question hit / answer miss) dominate full session miss?
- 59 • **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under
 60 measured index distortion, and what incremental gains does Soft O2
 61 provide over FlatIP and hard parent hydration?
- 62 • **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow
 63 (BIRCH) pathway effective? Does Soft O2-C help under standard same-
 64 session gold, and under a fair Mix of same-/cross-session gold?
- 65 • **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall
 66 channels, how large and session-tied are Soft O2 gains, and how do
 67 Soft O2 and Soft O2-C /Hybrid complement each other?

Contributions..

1. A dual-store memory architecture for Chinese legal consultation archives, separating fast episodic turn recovery from slower BIRCH semantic integration under same-domain interference.
2. Soft O2, a turn-level session soft-binding operator that addresses episode incompleteness without hard parent hydration or session-max score copy.
3. A conservative three-candidate diagnostic of Soft O2 score competition using the gap ratio $\rho = s_d/s_h$; default $\beta=0.98$ is chosen by validation sweep, not by a closed-form optimum.
4. An honest comparison with strong lexical baselines: BM25-joint and dense joint-episode indexing are highly competitive on exact/surface-overlap channels, while Soft O2 targets paraphrase, follow-up, and advice-recall under an online turn-level index.
5. Exploratory Soft O2-C / gated Hybrid results under a gold-assisted Mix solvability bound, plus corrected nDCG, failure taxonomy, and paired significance reporting.
6. O1 exact FlatIP and O3 bulk-load consolidation as implementation operators enabling the shared-store ablations reported here.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL operators and the Soft O2 diagnostic model. Section 4 details corpora, corrected metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM

103 improve statutory and advisory generation Yue et al. (2023); Huang et al.
 104 (2023); Yue et al. (2024), but they still need a reliable memory index when
 105 prior advice must be recovered from a growing same-domain log.

106 Legal consultation archives add a further difficulty: high lexical overlap
 107 among topically adjacent matters, with the professionally useful evidence
 108 unit being the full question-answer episode. This setting motivates *dual-*
 109 *store* memory architectures that keep fast episodic turn retrieval separate
 110 from slower semantic clustering, and session-aware soft binding operators
 111 that recover answer turns under same-domain interference without collapsing
 112 rank competition among unrelated candidates.

113 2.2. Dual-store memory and episodic retrieval

114 Dual-store designs distinguish fast episodic encoding from slower seman-
 115 tic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving
 116 et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dia-
 117 logue agents extend context via hierarchical paging, summary banks, graphs,
 118 and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Ras-
 119 mussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang
 120 et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved
 121 units as flat passages rather than *multi-turn consultation episodes* whose
 122 answer turn may rank below an unrelated question turn sharing legal vo-
 123 cabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA
 124 over personal conversation logs Wu et al. (2024); Maharana et al. (2024),
 125 while broader memory-agent surveys emphasize persistence beyond the con-
 126 text window Wu et al. (2025); Tan et al. (2025). Recent conversational
 127 RAG work stresses reuse of historical search evidence across turns Mo et al.
 128 (2026). BIMS-LEGAL positions its dual-store contribution at this intersec-
 129 tion: episodic turn embeddings with *sid* for same-session recovery (Soft O2),
 130 and BIRCH centroids with Soft O2-C for cross-session recovery when query
 131 and gold evidence occupy different sessions but share thematic structure
 132 Zhang et al. (1996). Soft O2 directly addresses the episodic retrieval chal-
 133 lenge that parent-document hydration and session-max aggregation handle
 134 via hard score copy: under dense same-domain interference, a question turn
 135 can dominate top- k while its answer sibling remains absent, producing fluent
 136 yet ungrounded downstream generation unless siblings enter under attenu-
 137 ated inherited scores.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH pathway.

We reuse CLS vocabulary as an architectural analogy for BIMS-LEGAL: turn embeddings with timestamps, roles, and session identifiers as episodic elements; Soft O2 as episode binding on the episodic pathway; BIRCH centroids with Soft O2-C as semantic integration on the slow pathway; and shared-store distractors as interference. The dual-store split is the systemic design; Soft O2 and Soft O2-C are the algorithmic mechanisms evaluated on Chinese legal consultation archives.

165 3. Method: BIMS-LEGAL

166 3.1. Architecture overview

167 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
168 tary episodic and semantic stores over the same consultation archive.

169 **Definition 1** (Episodic and semantic elements). *An episodic element is*
170 $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, *where* $\mathbf{v}_i \in \mathbb{R}^d$ *is an embedding,* t_i *a timestamp,* c_i
171 *conversational context (including speaker role),* ϕ_i *surface features, and* sid_i
172 *the consultation session identifier. A semantic element is* $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$
173 *with centroid* $\mathbf{c}_j$, *member set* C_j , *label* κ_j , *and formation statistics* τ_j . *Write*
174 $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$ *for session siblings and* $\text{Clus}(t) = \{t' : \text{cid}_{t'} =$
175 $\text{cid}_t\} \setminus \{t\}$ *for BIRCH-cluster siblings.*

176 **Definition 2** (Dual-store retrieval map). *Given query* q *with embedding*
177 $\mathbf{v}_q$, *base episodic retrieval returns a scored candidate set* $\mathcal{R}_0(q) = \{(t, s_t)\}$
178 *under Eq. (1). Soft O2 produces* $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ *(Sec-*
179 *tion 3.3). Soft O2-C produces* $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ *(Sec-*
180 *tion 3.5). Gated Hybrid composes session binding then cross-session cluster*
181 *binding (Algorithm 2). The returned list is the top-k of the bound set under*
182 *the rewritten scores.*

183 3.2. Base scoring and implementation operators (O1, O3)

184 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
185 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

186 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

187 with Δ_t the elapsed time between query time and the turn timestamp. Unless
188 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
189 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
190 and do not replace Eq. (1).

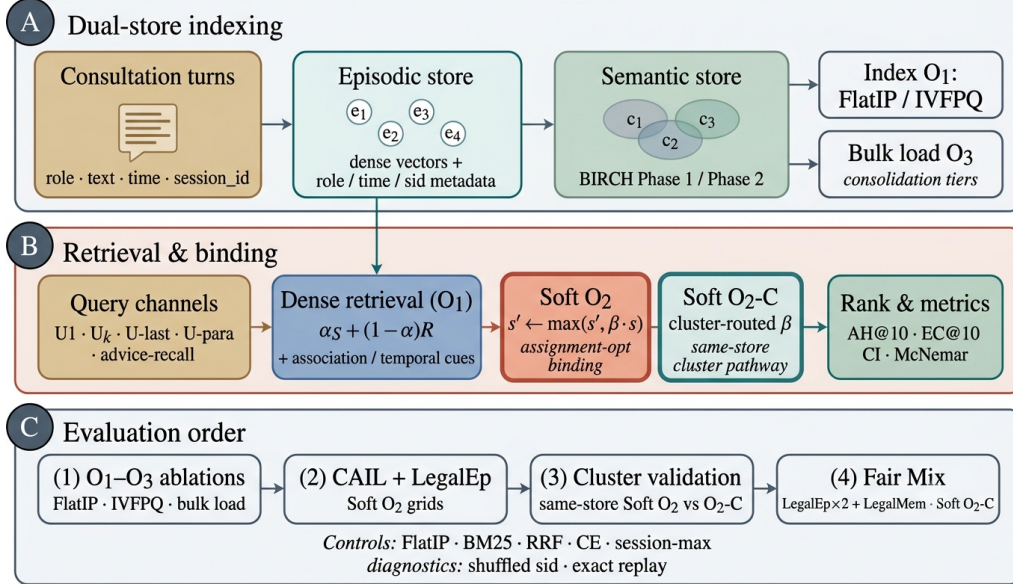


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load consolidation). (B) Query-time dense retrieval under O1 FlatIP with Soft O2 session binding and Soft O2-C cluster binding (algorithmic core); O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, CAIL /LegalEp Soft O2 grids, same-store cluster validation, and fair Mix Soft O2-C.

191 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 192 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 193 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 194 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 195 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 196 ble A.8: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 197 engineering prerequisite for rigorous Soft O2 evaluation, not a standalone
 198 retrieval algorithm.

199 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 200 insert update ($e_i \mapsto \text{BIRCH}(e_i); \text{Flush}$) by a deferred operator **Finalize** =
 201 $\text{BIRCH}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic writes. O3 does
 202 not change retrieval semantics; it makes $M \geq 400$ shared-store experiments
 203 runnable (≈ 11 min wall-clock for $M=400$; amortized ≈ 0.825 s/turn over
 204 ~ 800 turns). Quality-latency curves to $M=6400$ appear in Section 5.6.

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

205 *3.3. Soft O2: session soft binding*

206 *Motivation..* Under same-domain interference, a retrieved question turn can
207 score highly while its answer sibling remains outside top- k —episode incom-
208 pleteness compounded by IVFPQ score collapse and context fragmentation.
209 Soft O2 completes episodes without hard score copy.

210 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
211 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

212 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
213 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
214 *each member score by $\max_{t \in sid} s_t$.*

215 Algorithm 1 states the deployed Soft O2 procedure (expand + complete-
216 ness truncate), matching the public implementation.

217 *3.4. Soft O2 attenuation as diagnostic score competition*

218 The attenuation β controls how strongly a retrieved turn promotes its
219 session siblings. We treat the following as a *diagnostic* model, not a proof

of a universal optimum. In particular, a zero-margin hinge objective with $\beta \in (0, 1]$ does not identify a meaningful quantile solution: when the rank margin is zero, the rank term is inactive for all $\beta < 1$, so its weight cannot determine a closed-form optimum.

Definition 4 (Three-candidate diagnostic model). *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top- k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band). *Under Definition 4, assume the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

Proof. If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e. $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when $s_h > 0$. $\square$

Exploratory soft-rank diagnostic. We also minimize an exploratory soft pairwise diagnostic on a validation grid,

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

with logistic σ , temperature τ , and rank weight γ . This visualization is useful for studying the availability–ranking trade-off, but it depends on the diagnostic pool, query channel, and chosen hyperparameters. Default $\beta=0.98$

Table 2: Diagnostic gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). These are calibration diagnostics, not derivations of an optimal β .

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

is selected from the sweep in Table A.6, where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band.

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion; we report a single hard-expansion baseline. Dense joint episode indexing is an atomic-index bound with a different evidence unit.

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang et al. (1996).

Definition 5 (BIRCH dual-phase consolidation). *Let $\{\mathbf{c}_j\}$ be current centroids and θ_H, θ_M the assign /merge thresholds. **Phase 1 (assign)**. For a new embedding $\mathbf{v}$, set $j^* = \arg \max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^*}) \geq \theta_H$ then $C_{j^*} \leftarrow C_{j^*} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^*}$, else spawn a new cluster. **Phase 2 (merge)**. Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-duplicates above a redundancy threshold. Under O3, both phases run inside Finalize. Each turn t receives a cluster id cid_t , inducing $\text{Clus}(t)$ in Definition 1.*

Definition 6 (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default $\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large thematic clusters cannot flood top- k .*

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers cluster binding only on direct dense hits. This prevents thematic confusers from displacing same-session siblings already recovered by Soft O2. Soft O2 and Soft O2-C are complementary: Soft O2 is the default binder when query

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

- 1: $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$ $\triangleright$ session pathway first
- 2: $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$
- 3: $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$ $\triangleright$ direct dense hits only
- 4: **for** each trigger $t \in \mathcal{T}_{\text{trig}}$ **do**
- 5: $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$
- 6: **for** each $t' \in \text{Clus}(t)$ with $sid_{t'} \notin \mathcal{S}_{\text{cov}}$, up to budget B_c **do**
- 7: $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$
- 8: **end for**
- 9: **end for**
- 10: **return** top- k of $\mathcal{R}$

276 and gold share sid ; Soft O2-C is required when gold evidence is held in a
277 different session but the same cluster.

278 4. Corpora and Evaluation Protocol

279 Figure 2 sketches the shared-corpus design used throughout the legal eval-
280 uation. Gold needles and same-domain distractors occupy one store. Each
281 query probes that shared archive under fixed interference conditions. Pri-
282 vately rebuilt haystacks are avoided so that system comparisons isolate the
283 retrieval operator.

284 4.1. Datasets and query channels

285 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
286 Gold needles are authentic preliminary and final dialogues, with conversation
287 turns completed by label turns where applicable. Query channels cover the
288 first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user
289 turn (U-last), and constrained paraphrases (U-pa). Evidence defaults to
290 assistant turns in the target session.

291 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
292 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
293 a natural (question, answer) pair with session identifier sid . Reconstruction
294 applies length bounds and legal-content filters, excludes exam prompts, re-
295 moves near-duplicates, separates needles from distractors under a fixed seed,
296 and retains an audit subsample. After filtering, each LegalEp corpus retains

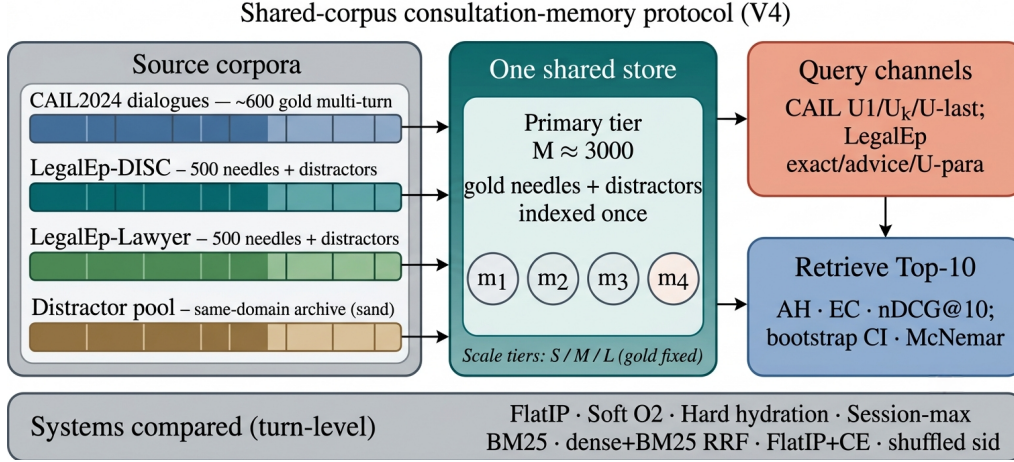


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary tier).

Query-channel policy (addressing exact-match shortcuts). Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary claims use paraphrase, U_k -followup / $U\text{-last}$, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$, $S=300$.

4.2. Metrics and baselines

Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC), and corrected normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is binary: whether a gold answer turn appears in the top- k . EC is the mean coverage of gold turns within the target session. Session Hit@ k is reported only for diagnosis. For nDCG we use fixed graded relevance over the

316 full gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
 317 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
 318 gold-relevant turns in the target session and truncated at k , so every system
 319 on a query shares the same IDCG. Earlier draft values based on retrieved-
 320 item IDCG are not comparable. We therefore treat Answer Hit and Episode
 321 Completeness as the primary numeric claims in the main Soft O2 grids, report
 322 a failure taxonomy derived from Session Hit versus Answer Hit, and release a
 323 corrected-nDCG recompute script (`paper/scripts/recompute_corrected_metrics.py`)
 324 with fixed-gold IDCG artifacts under `paper/ipm/figures/`. Primary con-
 325 trasts use bootstrap 95% CIs and McNemar mid- p tests on paired per-query
 326 hits; where families of related hypotheses are interpreted jointly, we also
 327 report Holm-corrected p -values (Table A.7).

328 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
 329 soft sibling score inheritance on the episodic pathway. Soft O2-C applies
 330 the analogous rule within BIRCH clusters. Hard parent hydration expands
 331 a hit to all siblings by copying the trigger score. Lexical controls include
 332 BM25 over turns and over joint question-answer documents, together with
 333 dense+BM25 Reciprocal Rank Fusion (RRF). A cross-encoder (CE) control
 334 reranks the top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`.
 335 Negative and sensitivity controls include shuffled *sid*, a β sweep, and IVFPQ
 336 as a quantization reference.

337 4.3. Fair Mix protocol for Soft O2-C

338 On fully same-session LegalMem grids, Soft O2 already recovers gold
 339 answers that share *sid* with the query, so Soft O2-C has little exclusive head-
 340 room and may inject thematic confusers. We therefore reconstruct the *same*
 341 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes
 342 are split into query-side S_a and evidence-side S_b with distinct session identi-
 343 fiers, while 30% remain intact. Distractors are unchanged. All operators—
 344 FlatIP, Soft O2, Soft O2-C, and gated Hybrid—share unrestricted dense re-
 345 trieval on this store; we report overall AH and stratum AH on `cross_session`
 346 vs `same_session` queries.

347 *Glue as a solvability bound (not a retrieval cheat)*.. After BIRCH consolida-
 348 tion on the Mix store, each cross-session S_a/S_b split pair is *glued* into one
 349 cluster identifier. This step uses gold split metadata available only in the
 350 evaluation protocol—it does *not* inject labels at query time and is *not* used
 351 in real unsupervised clustering deployments. Its sole purpose is to establish

352 a **solvability bound**: if Soft O2-C cannot recover cross-session gold when
 353 the semantic store *must* contain both halves of a split pair in one cluster,
 354 then failure reflects mechanism limits rather than accidental cluster separation.
 355 Glue therefore tests whether cluster soft binding *can* complete episodes
 356 when the slow pathway has the requisite thematic co-membership; natural
 357 same-cluster rates without glue are lower and are reported only as diagnos-
 358 tics (Section 6.3). Soft O2’s session membership constraint is unchanged—
 359 glue affects only whether Soft O2-C has a cluster in which to bind, not
 360 whether Soft O2 receives privileged label access. LegalEp Mix uses advice-
 361 recall queries; LegalMem Mix uses mid-split Uk-followup queries. External
 362 LongMemEval /LoCoMo numbers from prior retrieval runs on this codebase
 363 (QA correctness 70.7% / 68.2% on released test splits) are cited only as long-
 364 horizon memory context Wu et al. (2024); Maharana et al. (2024), not as
 365 Soft O2-C evidence on legal corpora.

366 5. Experiments

367 Unless noted, the embedding model is `nomic-embed-text-v2-moe` Nuss-
 368 baum et al. (2025), $k=10$, seed 42. We report primary contrasts in figures
 369 and keep full numeric grids in the appendix. Result files and scripts are re-
 370 leased with the accompanying repository so that each table maps to a public
 371 JSON artifact.

372 5.1. Implementation operators and Soft O2 ablation

373 Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA
 374 ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone
 375 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 376 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 377 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 378 over baseline) and Episode Completeness to 0.888 / 0.840—the primary al-
 379 gorithmic gain. O3 is enabled for all configurations in this ladder as a con-
 380 struction prerequisite; it does not alter retrieval scores.

381 5.2. RQ1 failure taxonomy

382 We classify each query into *complete* (answer evidence in top- k), *incom-*
 383 *plete* (target session partially retrieved but answer missing), or *session miss*
 384 (no target-session turn in top- k). Table 4 shows that incomplete retrieval is a
 385 substantial recurrent failure mode for FlatIP on primary non-exact channels,
 386 motivating Soft O2.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus.

Corpus	System	EC@10	AH@10	nDCG@10
DISC-Law	IVFPQ baseline	0.770	0.540	—
	+ O1 FlatIP	0.782	0.567	—
	+ O1+O2 Soft O2	0.888	0.780	—
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397	—
	+ O1 FlatIP	0.698	0.397	—
	+ O1+O2 Soft O2	0.840	0.680	—

Table 4: RQ1 failure taxonomy (FlatIP, $k=10$). LegalEp-DISC rows use corrected fixed-gold recompute; other rows use Session-Hit / Answer-Hit aggregates from the primary V4 grids.

Corpus / channel	System	Complete	Incomplete	Session miss
LegalEp-DISC / u-para	FlatIP	70.8%	26.8%	1.6%
LegalEp-DISC / advice	FlatIP	48.0%	17.0%	35.0%
LegalEp-Lawyer / u-para	FlatIP	71.5%	25.3%	0.8%
CAIL / Uk-followup	FlatIP	45.8%	54.0%	0.2%

387 5.3. Strong lexical and joint-episode baselines

388 BM25-joint indexes each question–answer episode as one lexical document
389 and is therefore very strong on exact replay and other high-overlap channels.
390 Soft O2 instead assumes an online turn-level index and uses session bind-
391 ing to recover missing answer turns after a dense hit. Table 5 reports this
392 trade-off on the same store, query set, and $k=10$. Soft O2 is not claimed
393 to dominate BM25-joint universally; its advantage is on non-exact channels
394 and on preserving turn-level ranking flexibility.

395 5.4. Soft O2 on session (CAIL and LegalEp)

396 Figure 3 summarizes AH@10 and EC@10 on authentic CAIL multi-turn
397 channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP
398 on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs
399 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200),
400 tying the improvement to correct episode membership (RQ1–RQ2). Hard
401 expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while
402 lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a

Table 5: BM25-joint / joint-QA vs Soft O2 (AH@10, primary tier). Joint baselines are strongest on exact/surface overlap; Soft O2 targets paraphrase and advice-recall.

Corpus / channel	BM25-joint	Joint QA	Soft O2
CAIL / U1	0.993	—	0.788
CAIL / Uk	0.840	—	0.698
CAIL / U-last	0.823	—	0.762
LegalEp-DISC / exact	0.976	—	0.638
LegalEp-DISC / u-para	0.592	—	0.648
LegalEp-DISC / advice	0.432	—	0.428
LegalEp-Lawyer / exact	—	—	0.752
LegalEp-Lawyer / u-para	—	—	0.759
LegalEp-Lawyer / advice	0.652	—	0.472

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

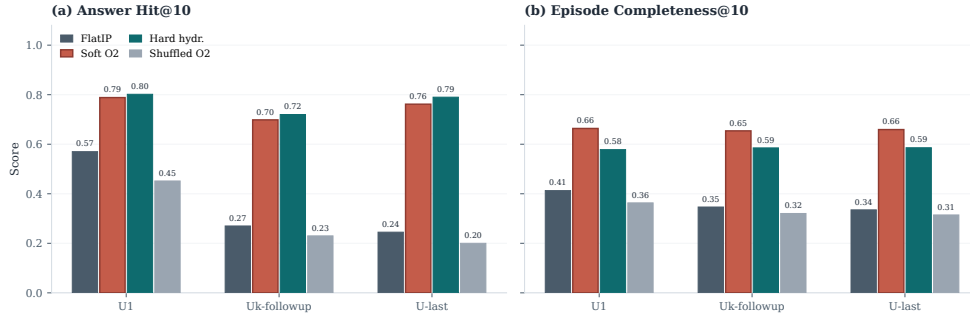


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

rank-preserving binder: siblings enter under attenuated scores, and episode
 coverage remains higher than under hard score copy. On CAIL, nDCG often
 falls relative to FlatIP even as AH and EC rise, reflecting an availability-
 ranking trade-off. Soft O2 still exceeds FlatIP+CE on every CAIL channel
 (Table A.5). Full rows appear in Table A.1.

Figure 4 reports LegalEp transfer. On LegalEp-DISC exact replay Soft O2
 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already
 recoverable for many queries—consistent with treating exact as diagnostic.
 On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNe-
 mar mid- $p=0.024^*$). Paraphrase yields the clearest LegalEp gains: Soft O2
 reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, ex-

Table 6: Corrected graded nDCG@10 (fixed-gold IDCG) on LegalEp and CAIL primary channels under a fresh FlatIP rebuild. Soft O2 reduces incompleteness while raising AH and corrected nDCG.

Corpus	Channel	System	AH@10	nDCG@10	Incomplete
LegalEp-DISC	u-para	FlatIP	0.716	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.895	0.759	8.7%
LegalEp-DISC	advice	FlatIP	0.480	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.608	0.524	3.8%
LegalEp-Lawyer	u-para	FlatIP	0.739	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.946	0.844	3.8%
LegalEp-Lawyer	advice	FlatIP	0.444	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.652	0.582	2.2%
CAIL	Uk-followup	FlatIP	0.458	0.445	54.0%
CAIL	Uk-followup	Soft O2	0.928	0.793	7.0%

ceeding hard expansion on both corpora (mid- $p < 0.001^{**}$ on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid- $p < 0.001^{**}$), while shuffled *sid* falls toward or below FlatIP. Detailed grids are in Tables A.2–A.3.

5.5. Same-store cluster pathway validation

Table 7 contrasts Soft O2 and ungated Soft O2-C on standard same-session LegalEp /LegalMem stores (unrestricted dense; gold answers share *sid* with the query). Soft O2 dominates Soft O2-C on every reported channel (e.g., LegalMem U1 AH 0.903 vs 0.712; LegalEp-DISC exact 0.836 vs 0.744). The slow pathway is therefore *not* a free upgrade when episode membership already encodes the gold answer: cluster siblings inject thematic confusers that displace session-complete evidence.

Table 8 reports the complementary fair Mix setting (RQ3–RQ4): the same stores with 70% cross-session Sa/Sb gold and 30% intact same-session gold, unrestricted dense for every operator. Gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer overall (AH 0.534 vs 0.496; mid- $p = 0.010^*$) and on the cross-session stratum (0.523 vs 0.446; mid- $p < 10^{-4^{**}}$). On LegalEp-DISC, Hybrid is directionally above Soft O2 (0.486 vs 0.482) but not significant at $N = 500$. On LegalMem-MT, Hybrid attains the highest overall AH (0.646 vs Soft O2 0.618; mid- $p = 0.18$; cross-session mid- $p = 0.095$) while preserving most same-session hits (0.867 vs 0.887). Ungated Soft O2-C

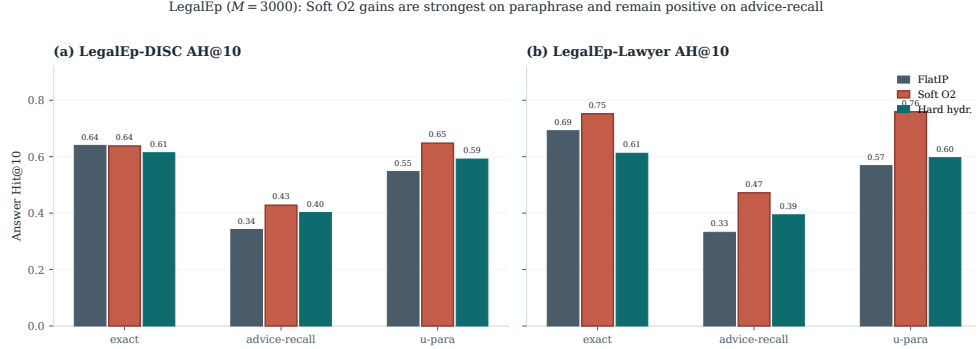


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Same-session Soft O2 vs Soft O2-C on the shared LegalEp /LegalMem stores (AH@10). Soft O2-C is ungated cluster soft binding. Boldface: better AH within each row.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

can raise cross-session hits but often displaces same-session siblings; Hybrid triggers cluster binding only on direct dense hits. Together, Tables 7–8 show that the BIRCH pathway is effective precisely when gold evidence is not co-session with the query.

McNemar mid- p tests on paired Answer Hit; * $p < 0.05$, ** $p < 0.01$.

5.6. Controls, significance, scale, and QA audit

Lexical BM25 is strong on exact replay (DISC exact BM25-turn 0.802, BM25-joint 0.976) and weaker on advice-recall (0.382 / 0.432). Dense+BM25 RRF complements Soft O2 on surface-overlap channels. Cross-encoder reranking (bge-reranker-v2-m3, top-30) improves FlatIP inside the CE suite, yet

Table 8: Fair Mix protocol (same store, unrestricted dense; AH@10). Gold is a 70%/30% mix of cross-session Sa/Sb splits and intact same-session episodes. Soft O2-C denotes ungated cluster binding; Hybrid is Soft O2+gated Soft O2-C. Boldface: best AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

remains below Soft O2 on CAIL (Uk FlatIP+CE 0.375 vs Soft O2 0.698; Table A.5). A β sweep peaks near $\{0.9, 0.95, 0.98\}$ and weakens at $\beta=0.5$ (Table A.6), consistent with the diagnostic gap-ratio band in Section 3.4 (Table 2). McNemar mid- p tests (Table A.7) show Soft O2 versus FlatIP highly significant on all CAIL channels and on LegalEp paraphrase /advice-recall.

Figure 5 and Table A.8 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary). A dual-protocol QA audit ($N=270$ per corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) links retrieval quality to downstream answerability under a *single* retrieval configuration (Soft O2); we do not claim causal superiority over FlatIP in generation without paired downstream controls. Full protocol and stratified results appear in Section 6.2. On an older revision-protocol store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

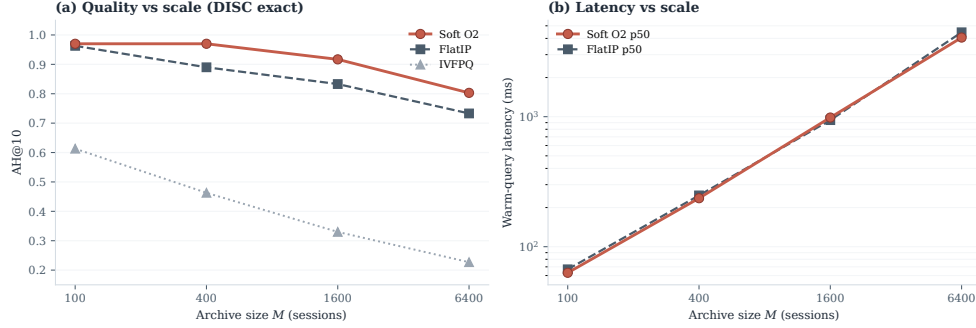


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

6. Discussion

6.1. Findings and scope

RQ1–RQ2 show that episode incompleteness is a substantial recurrent failure mode under same-domain interference (Table 4). O1 restores score fidelity as an implementation prerequisite; Soft O2 supplies the primary algorithmic gain through soft sibling inheritance; O3 enables archive construction at scale. The diagnostic gap-ratio analysis (Section 3.4) explains why β should remain close to but below unity, and why the validation sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. LegalEp paraphrase and advice-recall—not exact replay—carry the primary Soft O2 transfer claims. RQ3 shows that the BIRCH slow pathway benefits Soft O2-C only when gold is not already co-session; Mix Hybrid Soft O2+Soft O2-C recovers that regime without seed-forcing Soft O2 to zero. RQ4: Soft O2 (session) and Soft O2-C /Hybrid (cluster) are complementary operators within one BIMS-LEGAL dual-store. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019).

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this

end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder `legal_qa_n270`). Table 9 summarizes pooled Evidence-Answerability Rate (EAR) and stratified independent-judge scores. When Soft O2 retrieves complete episodes—both question and answer turns in top- k —the generator receives grounded prior advice and judged answerability rises in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60/\text{corpus}$) yields Cohen’s $\kappa=0.71$ / 0.68 and separates retrieval from generation failures (retrieval-failure share 0.283 / 0.317). Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). EAR: Evidence-Answerability Rate (judge score ≥ 0.7). Stratified judge scores: $n=90/\text{corpus}$ across exact, paraphrase, and follow-up protocols.

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL. Mix glue establishes a solvability bound for Soft O2-C evaluation (Section 4.3); natural same-cluster rates without glue are lower and are diagnostics only—real unsupervised BIRCH clustering would not receive split-pair metadata at build time. Soft O2 depends on correct session identifiers; Soft O2-C depends on cluster quality. The β diagnostic is

516 semi-empirical: it assumes a dominant distractor and a well-defined hit score,
 517 and the fitted ρ uses a stored user-turn embedding as an exact-channel query
 518 proxy; paraphrase /advice-recall gap laws may shift the feasible band slightly
 519 but leave the high- β operating region unchanged. LLM paraphrases need hu-
 520 man audit. Statutory correctness, citation validity, and professional liability
 521 remain unevaluated. Multiple channels inflate comparison multiplicity; we
 522 therefore report Holm-corrected p -values for the primary Soft O2 vs FlatIP
 523 family (Table A.7) alongside uncorrected McNemar mid- p tests.

524 *Latency and deployment scope..* Scale curves cover $M \in$
 525 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.8). At
 526 $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
 527 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
 528 should therefore be positioned as **high-precision offline or quasi-**
 529 **real-time consultation memory recovery**—e.g., pre-meeting case
 530 review, audit replay, or batch RAG over a firm’s advice archive—not as
 531 a sub-millisecond search engine. Future work will combine approximate
 532 nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C
 533 binding layered on top of faithful candidate retrieval, preserving episode
 534 completion while reducing query latency. Behavior at roughly 10^5 sessions is
 535 not measured, and we do not claim asymptotic scalability from O3 wall-clock
 536 alone.

537 7. Conclusion

538 BIMS-LEGAL targets consultation-memory retrieval under same-domain
 539 interference in Chinese legal archives through a dual-store design: an episodic
 540 turn index paired with a BIRCH semantic store. Soft O2 session soft bind-
 541 ing and Soft O2-C with gated Hybrid cluster binding are the algorithmic
 542 core, with Soft O2 attenuation β validated on a diagnostic gap-ratio model;
 543 O1 exact indexing and O3 bulk-load consolidation are implementation oper-
 544 ators that enable the shared-store ablations reported here. Under a shared-
 545 corpus protocol spanning CAIL multi-turn dialogues and rebuilt LegalEp
 546 episode corpora, Soft O2 improves Answer Hit over FlatIP on multi-turn,
 547 paraphrase, and advice-recall channels with collapsing shuffled-*sid* controls,
 548 and yields higher episode completeness than hard expansion on CAIL. Same-
 549 store Soft O2-C underperforms Soft O2 when gold shares *sid*; under fair
 550 Mix reconstructions, gated Hybrid Soft O2+Soft O2-C recovers cross-session

551 gold while Soft O2 remains strong on intact same-session gold. End-to-end
552 QA audit links complete episode retrieval to judged answerability and low
553 wrong-session contamination in downstream generation. Sparse, fusion, and
554 cross-encoder baselines situate Soft O2 among standard IR alternatives.

555 **Declarations**

556 *Funding*

557 Computational resources were provided by the Data Law Laboratory,
558 China University of Political Science and Law.

559 *Conflict of interest*

560 The authors declare no competing interests.

561 *Ethics approval*

562 This study uses publicly released research corpora and excludes confiden-
563 tial client data. The system is a research prototype for consultation-memory
564 retrieval.

565 *Data availability*

566 Code, processed evaluation corpora, and primary result files are publicly
567 available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory>. Upstream legal corpora: CAIL2024 consultation tracks;
568 Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_d
569 ata.
570 ata.

571 *Code availability*

572 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,
573 Soft O2-C), LegalMem/LegalEp pipelines, Mix builders, and table/figure
574 scripts are released in the same repository with JSON artifacts for each main
575 table.

576 *Author contributions*

577 L.X. (University of Winnipeg) designed BIMS-LEGAL and the
578 Soft O2 /Soft O2-C evaluation, implemented the system, ran the experi-
579 ments, and wrote the manuscript. L.H. (corresponding author; CUPL Data
580 Law Lab and Institute for Data Law) provided methodology guidance, revi-
581 sion, and supervision.

582 **References**

- 583 Ashley, K.D., 2017. Artificial Intelligence and Legal Analytics: New Tools
584 for Law Practice in the Digital Age. Cambridge University Press.
- 585 Askari, A., Aliannejadi, M., Abolghasemi, A., Kanoulas, E., Verberne, S.,
586 2024. Answer retrieval in legal community question answering, in: Ad-
587 vances in Information Retrieval (ECIR), Springer. pp. 477–485.
- 588 Atkinson, R.C., Shiffrin, R.M., 1971. The control of short-term memory.
589 Scientific American 225, 82–91.
- 590 Callan, J.P., Lu, Z., Xu, S., 1994. Passage-level evidence in document re-
591 trieval, in: Proceedings of the 17th Annual International ACM SIGIR
592 Conference on Research and Development in Information Retrieval, pp.
593 302–310.
- 594 Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z., 2024. M3-
595 embedding: Multi-linguality, multi-functionality, multi-granularity text
596 embeddings through self-knowledge distillation, in: Findings of the associ-
597 ation for computational linguistics: ACL 2024, pp. 2318–2335.
- 598 Cormack, G.V., Clarke, C.L.A., Buettcher, S., 2009. Reciprocal rank fusion
599 outperforms condorcet and individual rank learning methods, in: Proceed-
600 ings of the 32nd International ACM SIGIR Conference on Research and
601 Development in Information Retrieval, pp. 758–759.
- 602 Dai, Z., Xiong, C., Liu, Z., Callan, J., 2019. Context-aware sentence/passage
603 encoding for answering factoid questions, in: Proceedings of the 2019 Con-
604 ference on Empirical Methods in Natural Language Processing and the
605 9th International Joint Conference on Natural Language Processing, pp.
606 241–251.
- 607 Gutiérrez, B.J., Shu, Y., Gu, Y., Yasunaga, M., Su, Y., 2024. Hipporag:
608 Neurobiologically inspired long-term memory for large language models.
609 Advances in neural information processing systems 37, 59532–59569.
- 610 Huang, Q., Tao, M., Zhang, C., An, Z., Jiang, C., Chen, Z., Wu, Z., Feng,
611 Y., 2023. Lawyer llama technical report. [arXiv:2305.15062](https://arxiv.org/abs/2305.15062).
- 612 Johnson, J., Douze, M., Jégou, H., 2019. Billion-scale similarity search with
613 GPUs. IEEE Transactions on Big Data 7, 535–547.

614 Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen,
615 D., Yih, W.t., 2020. Dense passage retrieval for open-domain question
616 answering, in: Proceedings of the 2020 Conference on Empirical Methods
617 in Natural Language Processing (EMNLP), pp. 6769–6781.

618 Kumaran, D., Hassabis, D., McClelland, J.L., 2016. What learning systems
619 do intelligent agents need? complementary learning systems theory up-
620 dated. Trends in cognitive sciences 20, 512–534.

621 Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küt-
622 tler, H., Lewis, M., Yih, W.t., Rocktäschel, T., et al., 2020. Retrieval-
623 augmented generation for knowledge-intensive nlp tasks. Advances in neural
624 information processing systems 33, 9459–9474.

625 Liu, B., Wu, Y., Zhang, F., Liu, Y., Wang, Z., Li, C., Zhang, M., Ma, S., 2022.
626 Query generation and buffer mechanism: Towards a better conversational
627 agent for legal case retrieval. Information Processing & Management 59,
628 103051.

629 Louis, A., van Dijck, G., Spanakis, G., 2024. Interpretable long-form legal
630 question answering with retrieval-augmented large language models, in:
631 Proceedings of the AAAI Conference on Artificial Intelligence, pp. 22266–
632 22275.

633 Maharana, A., Lee, D.H., Tulyakov, S., Bansal, M., Barbieri, F., Fang, Y.,
634 2024. Evaluating very long-term conversational memory of llm agents, in:
635 Proceedings of the 62nd Annual Meeting of the Association for Computa-
636 tional Linguistics (Volume 1: Long Papers), pp. 13851–13870.

637 Martinez-Gil, J., 2023. A survey on legal question–answering systems. Com-
638 puter Science Review 48, 100552.

639 McClelland, J.L., McNaughton, B.L., O’Reilly, R.C., 1995. Why there are
640 complementary learning systems in the hippocampus and neocortex: in-
641 sights from the successes and failures of connectionist models of learning
642 and memory. Psychological review 102, 419.

643 Mo, F., Gao, Y., Wu, Z., Liu, X., Chen, P., Li, Z., Wang, Z., Li, X., Jiang,
644 M., Nie, J.Y., 2026. Leveraging historical information to boost retrieval-
645 augmented generation in conversations. Information Processing & Man-
646 agement 63, 104449.

- 647 Modarressi, A., Imani, A., Fayyaz, M., Schütze, H., 2023. Ret-llm: Towards
648 a general read-write memory for large language models. arXiv preprint
649 arXiv:2305.14322 .
- 650 Nogueira, R., Yang, W., Cho, K., Lin, J., 2019. Multi-stage document rank-
651 ing with BERT. arXiv:1910.14497.
- 652 Nussbaum, Z., et al., 2025. Nomic embed multimodal: Interleaved text, im-
653 age, and datasets for general-purpose vision-language embeddings. Model
654 card: **nomic-embed-text-v2-moe** (Nomic AI). Embedding model used for
655 turn encoding in this work.
- 656 Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S.G., Stoica, I., Gonzalez,
657 J.E., 2023. Memgpt: Towards llms as operating systems. arXiv preprint
658 arXiv:2310.08560 .
- 659 Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., Chalef, D., 2025. Zep:
660 A temporal knowledge graph architecture for agent memory. arXiv preprint
661 arXiv:2501.13956 .
- 662 Robertson, S., Zaragoza, H., 2009. The probabilistic relevance framework:
663 BM25 and beyond. Foundations and Trends in Information Retrieval 3,
664 333–389.
- 665 Tan, H., Zhang, Z., Ma, C., Chen, X., Dai, Q., Dong, Z., 2025. Mem-
666 bench: Towards more comprehensive evaluation on the memory of llm-
667 based agents, in: Findings of the Association for Computational Linguis-
668 tics: ACL 2025, pp. 19336–19352.
- 669 Tulving, E., et al., 1972. Episodic and semantic memory. Organization of
670 memory 1, 1.
- 671 Wang, Y., Gao, Y., Chen, X., Jiang, H., Li, S., Yang, J., Yin, Q., Li, Z.,
672 Li, X., Yin, B., et al., 2024. Memoryllm: Towards self-updatable large
673 language models. arXiv preprint arXiv:2402.04624 .
- 674 Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.W., Yu, D., 2024. Long-
675 memeval: Benchmarking chat assistants on long-term interactive memory.
676 arXiv preprint arXiv:2410.10813 .

677 Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R.,
678 Liu, Y., 2025. From human memory to ai memory: A survey on memory
679 mechanisms in the era of llms. [arXiv:2504.15965](#).

680 Yue, S., Chen, W., Wang, S., Li, B., Shen, C., Liu, S., Zhou, Y., Xiao, Y.,
681 Yun, S., Huang, X., Wei, Z., 2023. Disc-lawllm: Fine-tuning large language
682 models for intelligent legal services. [arXiv:2309.11325](#).

683 Yue, S., Liu, S., Zhou, Y., Shen, C., Wang, S., Xiao, Y., Li, B., Yun, S.,
684 Shen, X., Chen, W., et al., 2024. Lawllm: Intelligent legal system with
685 legal reasoning and verifiable retrieval, in: International Conference on
686 Database Systems for Advanced Applications, Springer. pp. 304–321.

687 Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R.,
688 Smola, A.J., 2017. Deep sets, in: Advances in Neural Information Process-
689 ing Systems.

690 Zhang, T., Ramakrishnan, R., Livny, M., 1996. BIRCH: an efficient data
691 clustering method for very large databases, in: Proceedings of the 1996
692 ACM SIGMOD International Conference on Management of Data, pp. 103–
693 114.

694 Zhang, Y., Zhai, S., Meng, Y., Bi, S., Chen, Y., Qi, G., 2024. Event is
695 more valuable than you think: Improving the similar legal case retrieval
696 via event knowledge. *Information Processing & Management* 61, 103729.

697 Zhong, W., Guo, L., Gao, Q., Ye, H., Wang, Y., 2024. Memorybank: En-
698 hancing large language models with long-term memory, in: Proceedings of
699 the AAAI Conference on Artificial Intelligence, pp. 19724–19731.

700 Appendix A. Numeric grids and controls

701 Main Soft O2 claims are visualized in Figures 3–5; O1–O2 and Mix
 702 Soft O2-C tables appear in the main text. Tables below retain full Soft O2
 703 numeric rows for reproducibility.

Table A.1: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 marked — pending fixed-gold IDCG recompute (Section 4); AH/EC unchanged.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	—	[0.530,0.610]
	Soft O2	0.788	0.664	—	[0.757,0.820]
	Hard hydr.	0.802	0.578	—	[0.768,0.835]
	Shuffled O2	0.452	0.363	—	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	—	[0.237,0.307]
	Soft O2	0.698	0.654	—	[0.660,0.735]
	Hard hydr.	0.720	0.585	—	[0.685,0.753]
	Shuffled O2	0.230	0.320	—	[0.198,0.262]
U-last	FlatIP	0.245	0.335	—	[0.212,0.280]
	Soft O2	0.762	0.659	—	[0.725,0.797]
	Hard hydr.	0.790	0.586	—	[0.757,0.822]
	Shuffled O2	0.200	0.315	—	[0.172,0.232]

704 [†]Legacy retrieved-item IDCG values removed from this grid; Soft O2 claims use
 705 AH/EC. Corrected nDCG script: `recompute_corrected_metrics.py`.

Table A.2: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	—
	Soft O2	0.638	0.780	—
	Hard hydr.	0.614	0.598	—
	Shuffled O2	0.448	0.645	—
advice-recall	FlatIP	0.342	0.438	—
	Soft O2	0.428	0.503	—
	Hard hydr.	0.402	0.405	—
	Shuffled O2	0.310	0.421	—
u-param	FlatIP	0.547	0.681	—
	Soft O2	0.648	0.735	—
	Hard hydr.	0.592	0.571	—
	Shuffled O2	0.469	0.610	—

Table A.3: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	—
	Soft O2	0.752	0.848	—
	Hard hydr.	0.612	0.615	—
	Shuffled O2	0.570	0.747	—
advice-recall	FlatIP	0.332	0.464	—
	Soft O2	0.472	0.540	—
	Hard hydr.	0.394	0.403	—
	Shuffled O2	0.348	0.467	—
u-param	FlatIP	0.568	0.718	—
	Soft O2	0.759	0.819	—
	Hard hydr.	0.596	0.594	—
	Shuffled O2	0.592	0.712	—

Table A.4: BM25 turn/joint and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported cells.

Corpus	Channel	BM25-turn	BM25-joint
LegalEp-DISC	exact	0.802	0.976
LegalEp-DISC	advice-recall	0.382	0.432
LegalEp-DISC	exact (dense RRF)	0.780	—
LegalEp-DISC	advice (dense RRF)	0.394	—
CAIL	U1 / Uk / U-last	0.748 / 0.482 / 0.413	0.993 / 0.840 / 0.823
LegalEp-Lawyer	exact / advice	0.784 / 0.460	0.998 / 0.652

Table A.5: Cross-encoder control (AH@10). FlatIP and FlatIP+CE are paired in the CE suite (**bge-reranker-v2-m3** over top-30 dense candidates). Boldface: best AH within each row. Soft O2 reproduces the primary evaluation; FlatIP here may differ slightly from primary FlatIP due to independent CE-suite re-indexing.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

Table A.6: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.7: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.8: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503